

Lab 3: Architectural Pattern Selection – Written Justification

Problem Statement : 22

Smart Parking Space Rental & License Validator

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Architecture Selection

Microservices Architecture for the Smart Parking Space Rental & License Validator System.

Comparison of Styles :

Style	Pros for this system	Cons for this system	Fit
Layered	Simple, clear separation of UI / business / data	Whole app scales together; one slow layer delays gate checks	Medium
Client-Server	Central control, easy data consistency	Single point of failure; gates, apps and wardens all depend on one server	Low
Microservices	Independent scaling, fault isolation, independent deployment	Operational complexity, network latency	High

Architectural Choice

Microservices: the system is split into small, independently deployable components – Reservation Manager, Plate Recognition, Billing & Rate Engine, Payment Service, Overstay Monitor, the Vehicle Owner App / Warden Console clients and a Database component – that talk only through defined interfaces (IReservation, IPlateValidation, IFeeCalculation, IPayment, IViolationAlerts, IBookingData, etc.).

Two Reasons for the Choice

1. Different parts of the parking system have very different load. Gate-camera plate scans and reservation look-ups spike at rush hour, while Billing and Warden alerts are light. Microservices let the city scale Plate Recognition and Reservation Manager alone, instead of replicating the whole application.

2. Failures and changes must stay local. If the Payment gateway is down or the tariff rules change (progressive hourly rates), only the Payment or Billing service is fixed or redeployed; vehicles can still enter and exit because plate validation and reservations keep running. Each service can also be replaced as long as it keeps the same interface.

Security Advantage

Isolation of sensitive data and a small attack surface: card and payment data is handled only inside the Payment Service (the rest of the system sees a token/receipt), and each service exposes only its own interface over authenticated HTTPS/gRPC. A compromise of the Warden Console or Vehicle Owner App therefore cannot directly read payment data or database tables, and vehicle-plate (personal) data is limited to the Plate Recognition and Reservation services.

Performance Benefit

The gate path (Plate Recognition fi IPlateValidation fi Reservation Manager) is a short, dedicated call chain that can be scaled horizontally and cached independently, helping meet NFR-001 (plate lookup under 300 ms) even at peak load. Slower work such as billing, payment and overstay checks runs in separate services so it never blocks the gate barrier.

Trade-off Acknowledged

Microservices add network latency and operational complexity; this is accepted because the city-scale, real-time and multi-actor nature of the system outweighs the extra deployment effort. Components are shown in UML dig with provided (ball) and required (socket) interfaces